1. Roadmap for proof of Vasconcelos's conjecture.

Aim: to prove every $z \in \pi^2(R)$ is nilpotent.

① $\pi^2(R) \cong \mathrm{Hom}_k(kX_1, k)$

$\cong \mathrm{Hom}_Q(QX_1, k)$

$\cong \mathrm{Hom}_Q(I, k)$ since $\partial X_1 = I$

$\cong \mathrm{Hom}_R(I/I^2, k)$

$z \in \pi^2(R) \iff z : I/I^2 \to k$

② Since I/I^2 of finite projective dim,

$$z : I/I^2 \to N = I/I^2 \to k$$

factorises through a fin. pdim. module.

$\Downarrow$ passes to 1st syzygies

$$RX_1 \to Syz_1(N) \to m$$

$\Downarrow$

actually gives a map $RX \to Syz_1(N)$

$\Downarrow$

By universal property of $\Omega_{Q[x]/Q}$, this gives a chain derivation:

$\gamma : Q[x] \to Syz_1(N)$, and this factorises $\theta_z : Q[x] \xrightarrow{\theta_z} Q[x] \xrightarrow{\pi} R$.

(See Thm 4.3.10 in BEJ.)

③ By Thm 4.3.5 ☆:

θ_z factorises through fin. pdim.

$\Rightarrow$ Resolve $\mathrm{Syz}_1(N)$ by a finite semifree A-mod., and finiteness forces a cohomologous derivation to θ_z vanishes on $k[X_{\geq n}]$

$\Rightarrow$ $-\mathrm{ad}(z)$ vanishes on $\pi^{\geq n}(R) \to \pi^{>n}(R)$ (Lem 4.3.3 : $\widetilde{\theta_z} = -\mathrm{ad}(z)$).

$\Rightarrow \mathrm{ad}(z)^n = 0$, z nilp.

2. Thm (Avramov-Halperin):

TFAE:

① R is C.I.

② Every elt. in $\pi^2(R)$ is central.

③ Every elt. in $\pi^2(R)$ is nilpotent.

① $\Rightarrow$ ② $\Rightarrow$ ③ is clear: by degree reasons ① $\Rightarrow$ ②; central $\Rightarrow$ nilp. gives ② $\Rightarrow$ ③.

Thus from this, we have I/I^2 fin. pdim. $\Rightarrow R$ is C.I.